

FDS/IDS Assignment: Predicting Cardiac Diseases in Children

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The main goal of this work is to apply feature importance methods to rank variables that are more predictive of cardiac pathologies. Before trying to achieve this goal, it may be necessary to preprocess the data. For example, data cleaning, data transformation, data normalization, as well as removal of irrelevant features (such as the ID), will be needed to help improving data quality.

In data cleaning, you need to look for **inconsistencies**, **noise**, **missing values**, **duplicate objects** etc. For example, you may look for different variable values that convey the same meaning (M or Masculine, where both mean the Masculine gender) or variables that contain spurious values (for example, impossible values for ages, dates etc), you may need to remove duplicate entries/objects, remove variables that contain too many missing values (for example, remove rows or columns with more than 50% missing values, if missing values have no meaning), or remove variables that have the same value for all objects.

Data transformation involves applying functions that will convert variables or objects from one space to another. Methods commonly used are smoothing (binning, clustering or regression, for example), aggregation, generalization (for example, to transform low level feature values into higher level feature values - age represented in higher level intervals), normalization (for example, $x_{new} = \frac{x - x_{min}}{x_{max} - x_{min}}$) and standardization $x_{new} = \frac{x - \mu}{\sigma}$ to make data fall into a small specified range, or feature construction (for example, creation of new attributes such as mass body index from height and weight). Be careful with outliers. If your data has too many outliers, normalization may push all your “normal” values to a very small interval. In that case, it is recommended to use standardization or using the interquartile range method to remove the outliers. Make sure that you really need to remove outliers. Make sure that outliers are not noise or impossible values.

After preprocessing, it is usual to proceed with univariate analysis (summaries, histograms, boxplots, mean, standard deviation, percentiles, interquartile range etc to measure data spread, centrality etc), bivariate analysis (correlations, for example Pearson, Spearman, Kendall etc), and multivariate analysis (regression, clustering, factoring-PCA, mutual information etc). These are useful to study relevant or redundant variables and help to perform a pre-selection of features. Note that strong correlations among variables mean that they are dependent in some way. When performing feature selection, usually (but not always, because it depends on the domain), if two or more variables are strongly correlated, it may mean that they equally contribute to the analysis and some of them may be discarded. For example, if height and weight are strongly correlated with BMI (Body Mass Index), you should use only BMI. Note that BMI, height and weight are not linearly correlated and, therefore, methods such as Pearson or Spearman may not capture that relationship. Methods of **multiple mutual information** perform feature selection by choosing only relevant and non-redundant variables.

Besides performing statistical analysis, we can also apply machine learning algorithms to extract **knowledge** from data and generate predictive models. Examples of such algorithms are Support Vector Machines, Decision Trees, Random Forests, Bayesian Networks, Ensemble methods,

Neural Networks etc.

There are several different libraries available out there that can perform all those tasks. Be aware that algorithms offer a wide range of parameter choices. You need to be careful with that choice, and need to understand what algorithm is implemented and what the results mean.

In order to assess the performance (quality-wise or quantity-wise) of each method on the same dataset, we need to choose some evaluation metric. This can be the error rate, the rate of correctly classified instances (CCI), sensitivity (Recall, True Positive Rate), specificity, precision, Receiver Operating Characteristic (ROC) curves, Area Under the (ROC) Curve, Precision-Recall curves etc. The metric to be used to assess performance will usually depend on the domain in hand and on the machine learning task. For example, if data is skewed (unbalanced number of objects per class), error rate may not be the adequate choice (why?). If the target variable is numerical (not the case of this dataset, metrics commonly used are Mean Absolute Error - MAE, Root Mean Squared Error - RMSE, Mean Squared Error - MSE etc).

Data Dictionary The data to be used for this work was collected in the Real Hospital Português (RHP), Brazil, anonymized and shipped to Portugal with the approval of the RHP Ethics Committee. The Ethics Committee of the University of Porto, Portugal, also approved the use of this data for academic studies. Entries consist of children between 0 and 19 years old, with or without a cardiac pathology (variable NORMAL X ANORMAL).

Table 1 gives a brief description of the variables.

Variable PPA is calculated from PA SISTOLICA and PA DIASTOLICA, according to cardiac clinical tables that relate the patient gender with age and blood pressures (systolic and diastolic).

In order to be able to improve the quality of your analysis, it is required that you search for intervals normally used in pediatric cardiology, for example, for variables IDADE and IMC.

What to deliver? A report containing:

1. Raw data
 - describe variables according to their types: interval-scaled, binary, nominal, ordinal, ratio-scaled. Be aware that there are specific methods suitable to each type of variable.
 - Preliminary analysis (summaries, histograms, boxplots, spread measures, density). These are interesting to be applied to the raw data to “uncover” inconsistencies, outliers, duplicates etc.
 - List of main changes needed to be performed with the raw data.
2. Preprocessed data
3. Basic description (summaries, histograms, boxplots, spread measures, density).
4. Analysis
 - Bivariate analysis (correlations, regression)

ID	anonymized patient ID	
Peso	patient weight	
Altura	patient height	
IMC	body mass index	
Atendimento	date of visit	
DN	birth date	
IDADE	age	
Convenio	health care insurance	
PULSOS	pulses	
PA SISTOLICA	systolic blood pressure	
PA DIASTOLICA	diastolic blood pressure	
PPA	result SBP/DBP	
NORMAL x ANORMAL	absence or presence of pathology	se for esta a variável como é dicotômica têm que usar a regressão logística, caso contrário, se for uma variável contínua têm de usar a regressão linear
B2	type of the second heart sound	
SOPRO	murmur type	
FC	cardiac frequency	
HDA1	history of disease 1	
HDA2	history of disease 2 (other history)	
SEXO	patient gender	
MOTIVO1	first reason for being forwarded to the cardiology clinic	
MOTIVO2	second reason for being forwarded to the cardiology clinic	

Table 1: Data Dictionary

- Multivariate analysis (**multiple variable regression**, mutual information, **cluster analysis** - once more, be aware that some methods used to calculate similarity (dissimilarity) depend on the type of the variable. In the context of this dataset, when performing cluster analysis we are interested to know if there are groups of similar patients.
- Predictive Models (use of supervised **machine learning**)
 - Decision Tree learning
 - Support Vector Machines ver quais são os parametros de cada tipo de modelo e fazer com cross validation
 - Bayesian Networks
 - Others (if time allows)
- **Comparison**

5. Discussion and Main Conclusions

This work is to be performed by groups of at most two people.